One-Copy-Per-Sector Scope Restriction in the Fundamental Theorem Formalization

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1 The source argument

Cirac–Pérez-García–Schuch–Verstraete 2016 states the general proportional Fundamental Theorem as Corollary `II_cor2` of [CPGSV16]. The theorem applies to two tensors A and B whose matrix-product vectors are proportional, each presented in its basis-of-normal-tensor (BNT) canonical form. The canonical form for A is a direct-sum decomposition

$$A^i = \bigoplus_{j=1}^{g_A} I_{r_{A,j}} \otimes \mu_{A,j}(A^i_{j,1} \oplus \cdots \oplus A^i_{j,r_{A,j}}),$$

where each modulus class j contains $r_{A,j} \geq 1$ representative blocks $A_{j,1}, \ldots, A_{j,r_{A,j}}$ all sharing the same weight modulus $|\mu_{A,j}|$, and the moduli are strictly ordered across classes: $|\mu_{A,1}| > |\mu_{A,2}| > \cdots > |\mu_{A,g_A}|$. The same structure applies to B with multiplicity counts $r_{B,j}$.

The proof of Corollary `II_cor2` in [CPGSV16] proceeds in several steps. Step 3 (lines 1156–1163 of the arXiv version) derives the blockwise power-sum identity

$$\sum_{q=1}^{r_{A,j}} \mu^N_{A,j,q} = \sum_{q=1}^{r_{B,j}} (\nu_{B,j,q}\, e^{i\phi_j})^N, \qquad \text{for all } j \text{ and all } N \geq 1, \tag{1}$$

from the BNT linear-independence input together with the equal-MPV hypothesis (so the equality of matrix-product vectors forces the equality of the sums of Nth powers). Step 4 invokes Lemma `Lem:app_simple` of the same paper: if two finite sequences $(\lambda_{a,k})_{k=1}^{x_a}$ and $(\lambda_{b,k})_{k=1}^{x_b}$ satisfy

$$\sum_{k=1}^{x_a} \lambda^N_{a,k} = \sum_{k=1}^{x_b} \lambda^N_{b,k} \qquad \text{for } N = 1, \ldots, \max(x_a, x_b),$$

then $x_a = x_b$ and the two sequences are equal as multisets. Applied to (1), this recovers $r_{A,j} = r_{B,j} =: r_j$ and the multiset equality $\{\mu_{A,j,q}\} = \{e^{i\phi_j}\nu_{B,j,q}\}$ for each modulus class j. Step 6 then assembles the global gauge isomorphism as $Y = \bigoplus_j I_{r_j} \otimes Y_j$, using the per-class dimension matching $r_{A,j} = r_{B,j}$ from Step 4.

The mathematical structure of this argument is that the multiplicities r_j and the weight multisets are recovered from a finite family of polynomial identities in N, without any passage to a limit.

2 The one-copy-per-sector restriction

The surviving formal one-copy canonical form structure[1] imposes *strict* decreasing moduli across the entire index set: every block has a distinct modulus. This is a stronger condition than the source structure, which only requires strict ordering of the modulus *classes* while allowing $r_j \geq 1$ representatives within each class. The effect is to force $r_j = 1$ for every modulus class j: the formal canonical forms are restricted to the one-copy-per-sector special case.

In this special case, the source decomposition reduces to $A^i = \bigoplus_{j=1}^{g} \mu_j A_j^i$ with a single block per weight, and the general multiplicity structure $(r_{j,1}, \ldots, r_{j,r_j})$ collapses to a single representative. The identity (1) becomes the single-term equality $\mu_{A,j}^N = (e^{i\phi_j} \nu_{B,j})^N$, Steps 3–4 of the source proof become vacuous (there is nothing to separate), and the global gauge in Step 6 takes the simpler form $Y = \bigoplus_j Y_j$ (no identity tensor factor). The formalization thus covers only the one-copy-per-sector instance of Corollary `II_cor2`.

3 Missing components for the general case

Four items are currently absent from the formalization; each is required to lift the scope from $r_j = 1$ to $r_j \geq 1$.

Lemma `Lem:app_simple` (finite Newton–Girard uniqueness). The source paper states, in its appendix, that two finite sequences whose power sums agree for $N = 1, \ldots, \max(x_a, x_b)$ must be equal as multisets. This is a finite algebraic fact, provable by Newton–Girard identities: the power sums determine the elementary symmetric polynomials, which determine the multiset of roots. A module named ScalarPowerSumIdentity[2] was added at an early stage of the project, but it does not specialize to this particular fixed-length statement in the form needed by Step 4 of the source proof. The lemma needs to be formalized as a standalone result, separate from any BNT context.

BNT power-sum identity. The identity (1) requires a proof: starting from the BNT structure on A and B and the equal-MPV hypothesis, one must derive, for each modulus class j, the equality of the two power sums in (1). The formal development no longer packages coefficient arrays for the discarded proportional branch as standalone hypotheses. The source derivation of (1)

[1] `TNLean/MPS/BNT/Construction.lean`: IsNormalCanonicalFormBNT. The earlier non-normal wrapper was retired after its downstream consumers were replaced by explicit separated-block hypotheses.

[2] Introduced in commit `dd54b79e`.

from the BNT linear-independence data and the equal-MPV condition is still not carried out. This derivation is logically prior to applying `Lem:app_simple`.

Multi-copy sector structure. The surviving one-copy structure IsNormalCanonicalFormBNT, and any future non-normal analogue stated with explicit separated-block hypotheses, would need a field allowing equal-modulus groups: rather than a global strict ordering on the full index set, the condition should be a strict ordering on the set of modulus classes, with each class admitting an arbitrary finite number of representative blocks. Relaxing mu_strict_anti to this weaker form is the necessary type-level change. Without it, the general-multiplicity versions of Corollary `II_cor2` and of the gauge assembly step cannot be stated.

Global gauge assembly. With $r_j \geq 1$ copies per sector, the global gauge isomorphism takes the form $Y = \bigoplus_j I_{r_j} \otimes Y_j$, where I_{r_j} is the identity on the multiplicity space and Y_j is the per-block gauge on the bond dimension. Constructing and verifying this more general gauge requires the dimension match $r_{A,j} = r_{B,j}$ from Step 4, which in turn requires `Lem:app_simple` and the power-sum identity. The simpler $Y = \bigoplus_j Y_j$ gauge used in the one-copy case is already present.

4 What the current formalization does prove

Within the one-copy-per-sector restriction, the formalization establishes the equal-MPV branch of the Fundamental Theorem argument. The BNT separation hypotheses yield cross-block overlap decay, and the equal-MPV hypothesis at a single witness length is used to match the per-block gauge. The former one-shot proportional branch has been deleted: the surviving block-matching boundary uses the explicit conclusion BlockPermutationGaugePhaseConclusion only where the BNT comparison has already been supplied. Thus no current theorem is presented as deriving the source proportional comparison from the discarded coefficient hypotheses.

The source normalization $|\mu_1| = 1$ is now recorded as the field mu_dom_norm_one in IsNormalCanonicalFormBNT,[3] which brings the dominant-block normalization into alignment with the source convention. Under this normalization and the strict modulus ordering, every sub-dominant block has $|\mu_j| < 1$.

5 Downstream audit status

The downstream CPSV16 surfaces that currently consume the restricted normal-CF-BNT predicate have been audited against the one-copy-per-sector boundary.[4] Each of these public surfaces either assumes or constructs an

[3] IsNormalCanonicalFormBNT.mu_dom_norm_one, `TNLean/MPS/BNT/Construction.lean` lines 192–198.

[4] The relevant Lean surfaces are IsNormalCanonicalFormBNT, IsNormalCanonicalFormBNT.ofSeparatedData, IsNormalCanonicalFormBNT.cross_overlap_tendsto_zero, IsNormalCanonicalFormBNT.isBNT, IsNormalCanonicalFormBNT.exists_common_blockTensor_isInjective,

already chosen representative family, strictly ordered by weight modulus. They do not introduce a source-faithful canonical-form existence theorem and do not recover the repeated-copy sector data of the source BNT decomposition. Their docstrings and the corresponding Wielandt blueprint corollary therefore carry the same one-copy-per-sector marker.

This audit does not apply that marker to the sector-BNT development based on IsBNTCanonicalForm. That predicate is the paper-faithful sector surface used for the CPSV16 coefficient and weight-multiplicity arguments, where repeated copies inside one sector remain explicit.

6 Conclusion

The present formalization proves the proportional Fundamental Theorem in the one-copy-per-sector special case ($r_j = 1$ for every modulus class j). The general case of Corollary `II_cor2` in [CPGSV16], which allows $r_j \geq 1$, requires three additional components not yet in the formalization: the standalone Newton–Girard uniqueness lemma (`Lem:app_simple`), the derivation of the blockwise power-sum identity from the BNT structure and the equal-MPV hypothesis, and the global gauge assembly with multiplicity identity factors. Supplying these components would also require relaxing the mu˙strict˙anti field from a global strict ordering to a class-level ordering.

The corresponding tracked items are issue #1560 for the standalone `Lem:app_simple` lemma, issue #1561 for the BNT power-sum identity derivation, and issue #1562 for the general multi-copy route tracking the full relaxation.

References

[CPGSV16] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product density operators: Renormalization fixed points and boundary theories. Preprint; published version: Annals of Physics 378 (2017), 100–149, 2016.

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